

Mohammed Yasser Ismael

AI & DATA SCIENCE STUDENT

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About Me

AI & Data Science student with hands-on experience in Python, machine learning, NLP, and RAG-based applications. Skilled in developing ML pipelines, working with data, and building AI solutions using tools such as Scikit-learn, XGBoost, LightGBM, and Hugging Face. Interested in pursuing opportunities in Data Science, NLP, and AI Engineering.

Education

Faculty of Computers & Artificial Intelligence - Cairo University 2023 – 2027

Skills

Technical Skills:

- **Python** (Pandas, NumPy, Scikit-learn, Matplotlib)
- **SQL** (PostgreSQL, MySQL, SQL Server)
- **Data Analysis & Visualization** (Power BI, Excel)
- **Machine Learning** (Regression, Classification, Clustering, Feature Engineering)
- **Deep Learning** (CNN, RNN, Transformers, RAG, LLMs, Hugging Face, Fine-Tuning)

Soft Skills:

- Time Management
- Team Collaboration
- Adaptability
- Problem Solving

Work Experience

Summer Intern – CIB Bank July 2026

- Gained exposure to banking operations and the role of AI, data, and analytics in financial services.
- Developed an understanding of how data-driven technologies are applied to support banking processes and decision-making.

Data Analyst Intern – Uneeq Interns Oct 2025

- Analyzed 3 business datasets using Python, uncovering trends and actionable insights through data cleaning and exploratory analysis.

Data Analyst Intern – Elevvo Pathways Sep 2025

- Cleaned, analyzed, and visualized 4 datasets using Python and Excel, delivering business insights through interactive dashboards.

Projects

Arabic Legal AI Assistant (Graduation Project – National Telecommunication Institute)

- Collaborated in a team of 5 to develop an Arabic legal document question-answering assistant using a Retrieval-Augmented Generation (RAG) architecture.
- Handled document preprocessing and text preparation, including PDF text extraction and section-aware chunking of legal documents.
- Implemented Arabic text embeddings using BGE-M3 and integrated them with FAISS for semantic retrieval of relevant contract sections.
- Contributed to the retrieval pipeline that supplies relevant contract context to Google Gemini API for grounded question answering.

Loan Defaulter Prediction System (Graduation Project – Digital Egypt Pioneers Initiative)

- Led a team of 5 in developing an end-to-end loan default prediction system using 350K customer records and 150 features.
- Performed data preprocessing, feature engineering, class imbalance handling (SMOTE), and model development.
- Evaluated LightGBM, CatBoost, and Random Forest models, with LightGBM achieving the best performance.
- Deployed the final model as a web application for real-time credit risk assessment.

Energy Consumption Forecasting System

- Built a time-series forecasting pipeline using historical energy consumption data from 8 utility companies.
- Engineered temporal and lag-based features and trained an XGBoost model achieving $R^2 \approx 0.999$.
- Implemented recursive forecasting for future energy demand prediction.
- Deployed the solution as an interactive Streamlit application.

Supply Chain Latency & Risk Prediction

- Developed a machine learning pipeline on 180K+ shipment records to predict delivery outcomes across 4 classes.
- Applied dimensionality reduction and clustering techniques (PCA, UMAP, K-Means) for exploratory analysis.
- Addressed severe class imbalance using SMOTE and undersampling techniques.
- Tuned XGBoost and LightGBM models, achieving 73.7% accuracy and 0.85 F1-score for late-delivery prediction.

Business Insights & Executive Report for Olist

- Analyzed a multi-table e-commerce dataset covering customers, products, sellers, and orders.
- Evaluated customer behavior, product performance, and seller efficiency using Python and Excel.
- Delivered an executive-level business report with actionable recommendations to improve sales performance.

Courses

Natural Language Processing Summer Training – National Telecommunication Institute (NTI)

Data Scientist (Round 3) - Digital Egypt Pioneers Initiative (DEPI)

IBM AI Engineering Professional Certificate - IBM

Machine Learning Specialization – DeepLearning.AI

Data Scientist in Python – DataCamp

Data Analyst in SQL – DataCamp

Data Analyst in PowerBI – DataCamp